

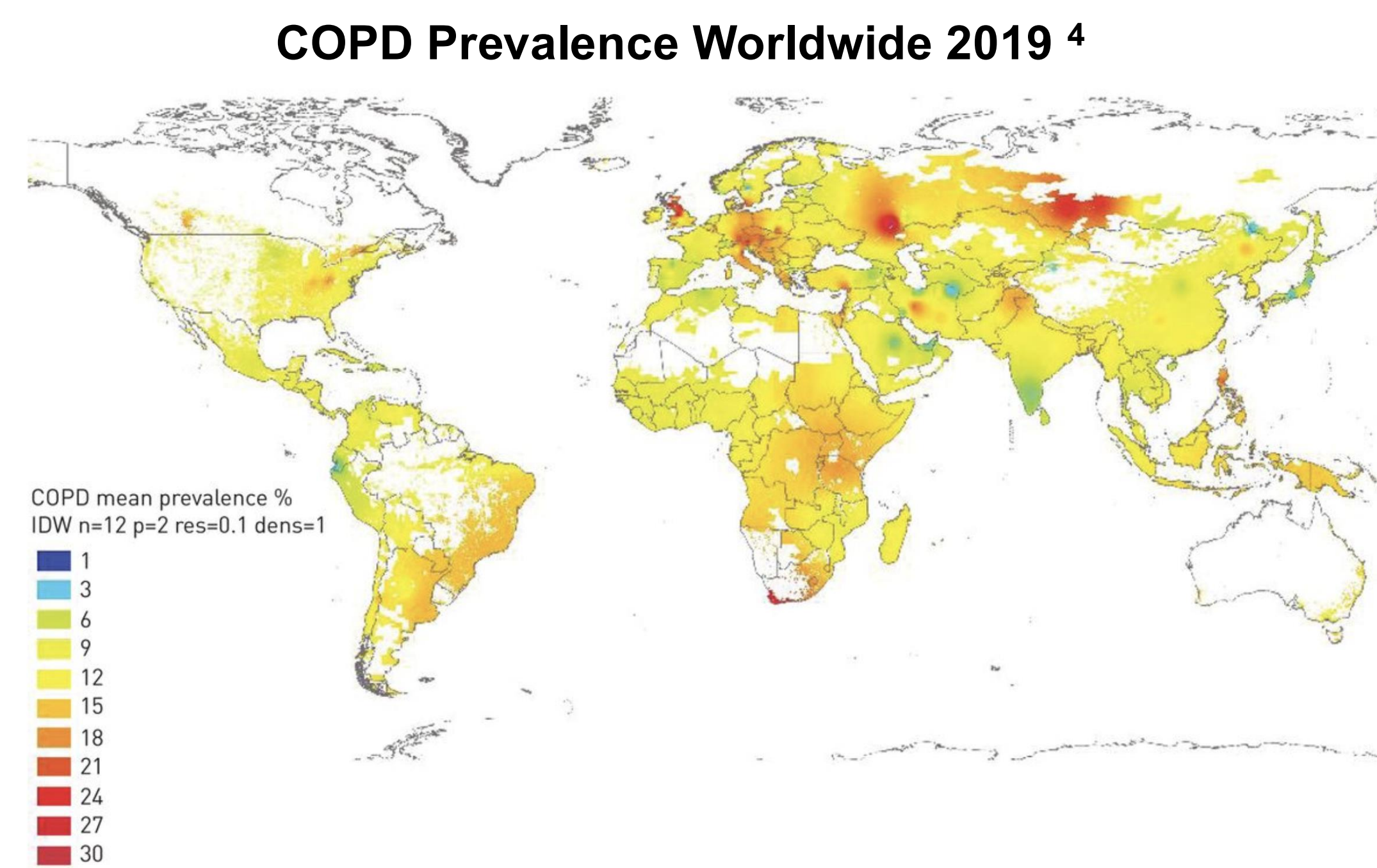
M. Yang,¹ N. M. Robertson,¹ A. K. Sharma,² R. K. Chandyo,³ L. Shrestha,³ S. K. Das,⁴ B. Kirenga,⁵ P. Alupo,⁵ G. E. Gianella,⁶ T. Siddharthan,⁷ S. L. Pollard,¹ S. Quaderi,⁸ N. Rykiel,¹ O. Flores-Flores,⁹ J. R. Hurst,⁸ R.A. Wise,¹ W. Checkley¹

¹ Division of Pulmonary and Critical Care, School of Medicine, Johns Hopkins University ² Department of Pediatrics and the Child Health Research Project, Tribhuvan University ³ Department of Community Medicine, Kathmandu Medical College ⁴ Division of Pulmonary and Critical Care, Tribhuvan University ⁵ Makerere University, Makerere Lung Institute ⁶ Universidad Peruana Cayetano Heredia, Facultad de Medicina, ⁷ Pulmonary & Critical Care, University of Miami, ⁸ UCL Respiratory, University College London, London ⁹ Universidad de San Martin de Porres, Lima, Peru

Introduction

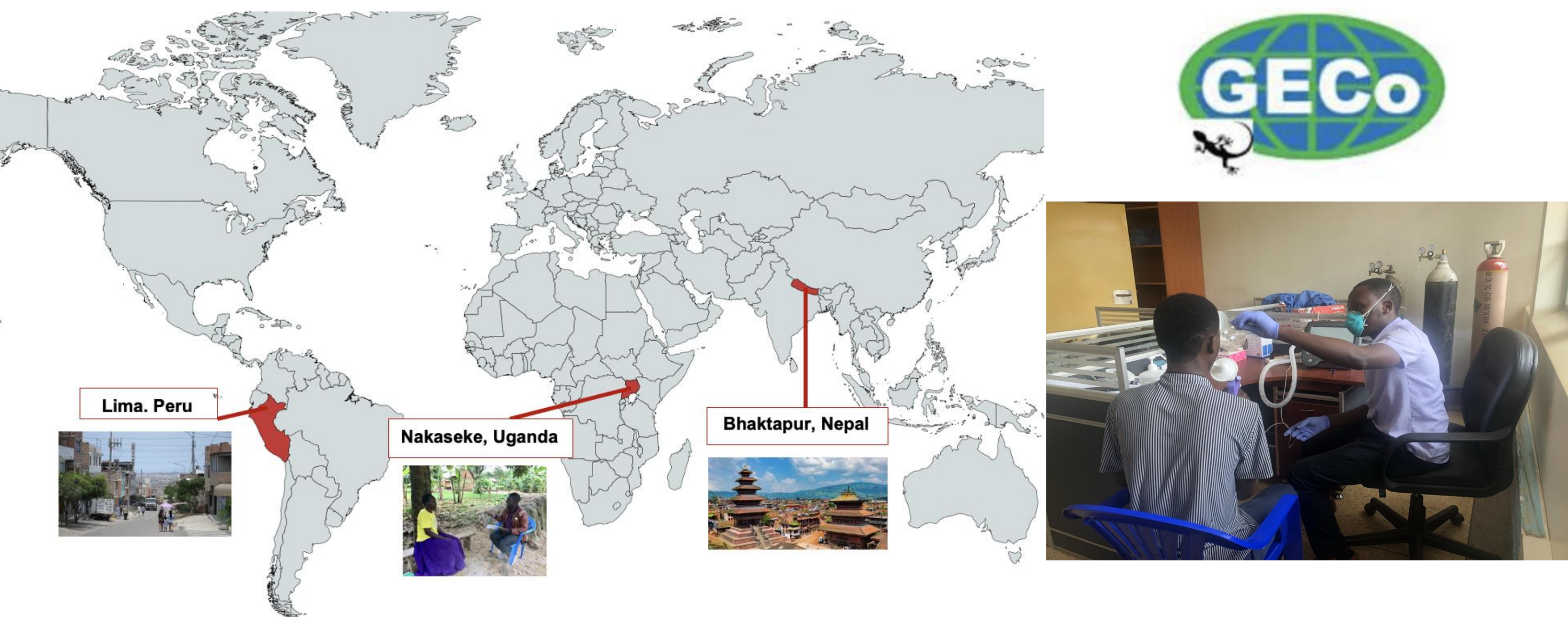
- COPD is a leading cause of morbidity and mortality globally ¹
- Access to spirometry is limited and costly ²
- Burden is likely underestimated due to limited access to spirometry and trained healthcare workers to perform spirometry ²
- Screening questionnaires may bridge the diagnostic gap ³

Aim: Evaluate performance Questionnaire of the St. George's Respiratory as a screening tool using spirometry-confirmed COPD as a gold standard.



Methods

- Global Excellence in COPD Outcomes (GECO) study screened 10,709 adults ≥ 40 years for COPD in Nepal, Peru, and Uganda 2018 - 2020
- Collected SGRQ and pre-bronchodilator peak expiratory flow (PEF) with handheld easy-to-use Piko-1 device
- COPD defined as post-bronchodilator FEV₁/FVC Z-score < -1.645
- Evaluated discriminative performance of SGRQ using ROC curves and identified the best threshold to screen for spirometry-confirmed COPD
- SGRQ's performance was combined with sex-specific PEF improved applicability across both sexes



Participant Demographics

- We included 10,008 participants and 9.5% had spirometry-confirmed prevalence of COPD. The mean (± SD) total SGRQ score was 7.9 ± 11.9 points
- There was significant difference in current biomass use between groups

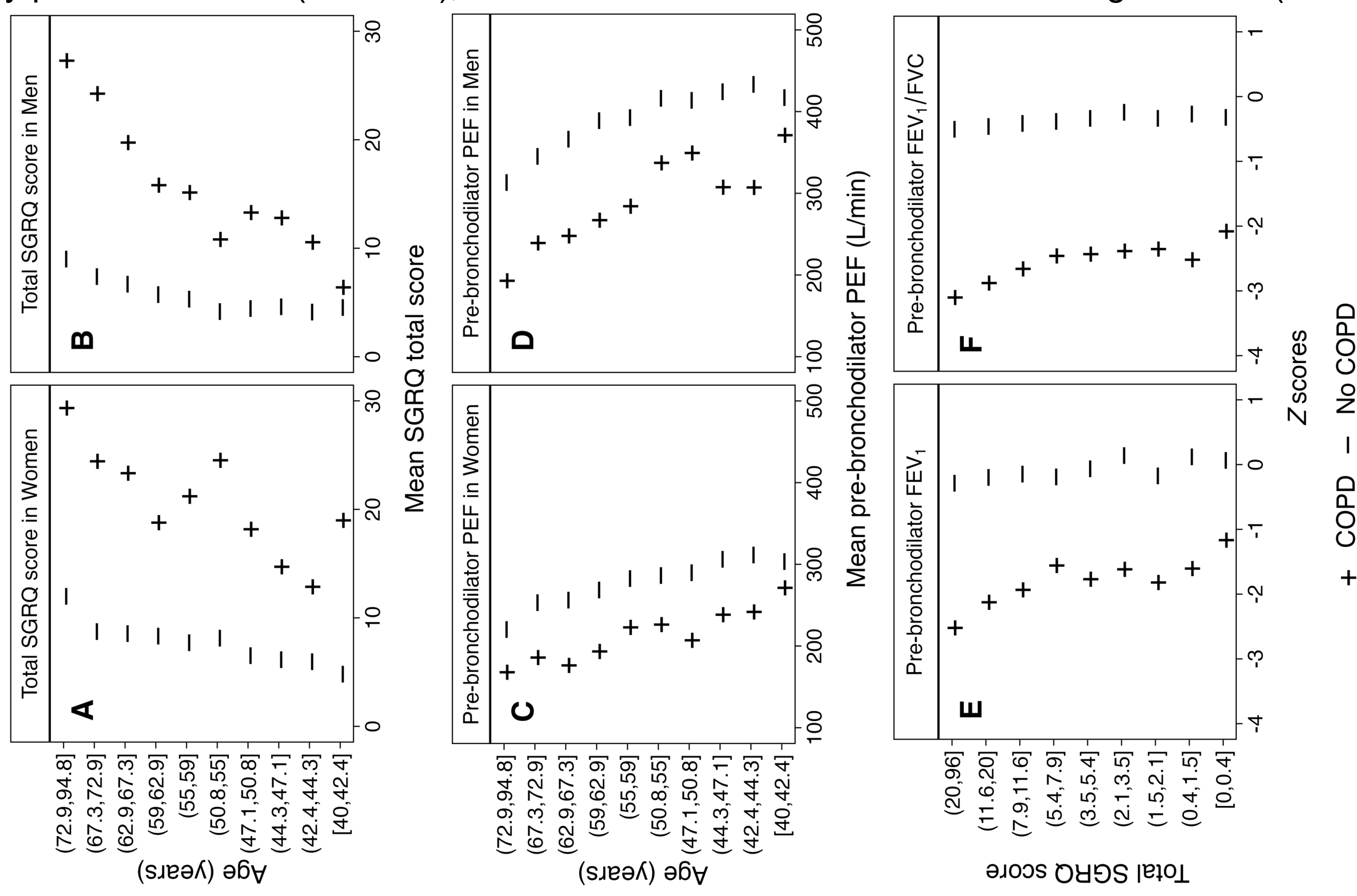
Table 1. Participant characteristics by COPD status

	COPD (n=948)	Non-COPD (n=9060)	P-value
Age in years, n (%)			
40 – 44	105 (11.1)	2273 (25.1)	<0.001
45 – 54	145 (15.2)	2456 (27.1)	
55 – 64	268 (28.3)	2224 (24.5)	
65 – 95	430 (45.4)	2107 (23.3)	
Mean age in years, mean (SD)	62.6 (11.8)	55.6 (11.5)	<0.001
Sex, n (%)			
Males	553 (58.3)	4402 (48.6)	<0.001
Females	395 (41.7)	4658 (51.4)	
SGRQ, mean (SD)			
Total	20.3 (19.4)	6.6 (9.9)	<0.001
Symptoms	24.1 (19.4)	11.6 (12.8)	<0.001
Impact	13.1 (19.6)	3.4 (9.0)	<0.001
Activity	31.0 (26.0)	9.7 (16.0)	<0.001
Pre-bronchodilator peak expiratory flow (PEF) in L/min, mean (SD)			
PEF by Piko-1	236.8 (95.4)	334.7 (113.2)	<0.001
PEF by Spirometry	255.0 (109.6)	418.1 (122.8)	<0.001
Smoking status, n (%)			
Never smoker	473 (49.9)	6003 (66.3)	<0.001
Former smoker	193 (20.4)	1777 (19.6)	
Current smoker	281 (29.7)	1279 (14.1)	
Current biomass fuel use, n (%)			
Yes	243 (25.6)	2585 (28.6)	0.06
No	705 (74.4)	6458 (71.4)	
Years of education, mean (SD)	6.9 (3.7)	7.9 (3.8)	<0.001
Wealth index, mean (SD)	3.2 (21.3)	-0.1 (16.3)	<0.001

Determinants of SGRQ and pre-bronchodilator lung function

- Age and sex were important determinants of the SGRQ score and pre-bronchodilator PEF.
- Mean total SGRQ score was 12.7 points (95% CI 11.9-13.4) higher in participants with COPD than in those without COPD, 2.2 points (95% CI 1.8-2.6) higher in men than in women, and 1.7 points (95% CI 1.5-1.9) higher for each older decade of age.
- The SGRQ was also associated with lower pre-bronchodilator forced expiratory volumes in participants with and without COPD.

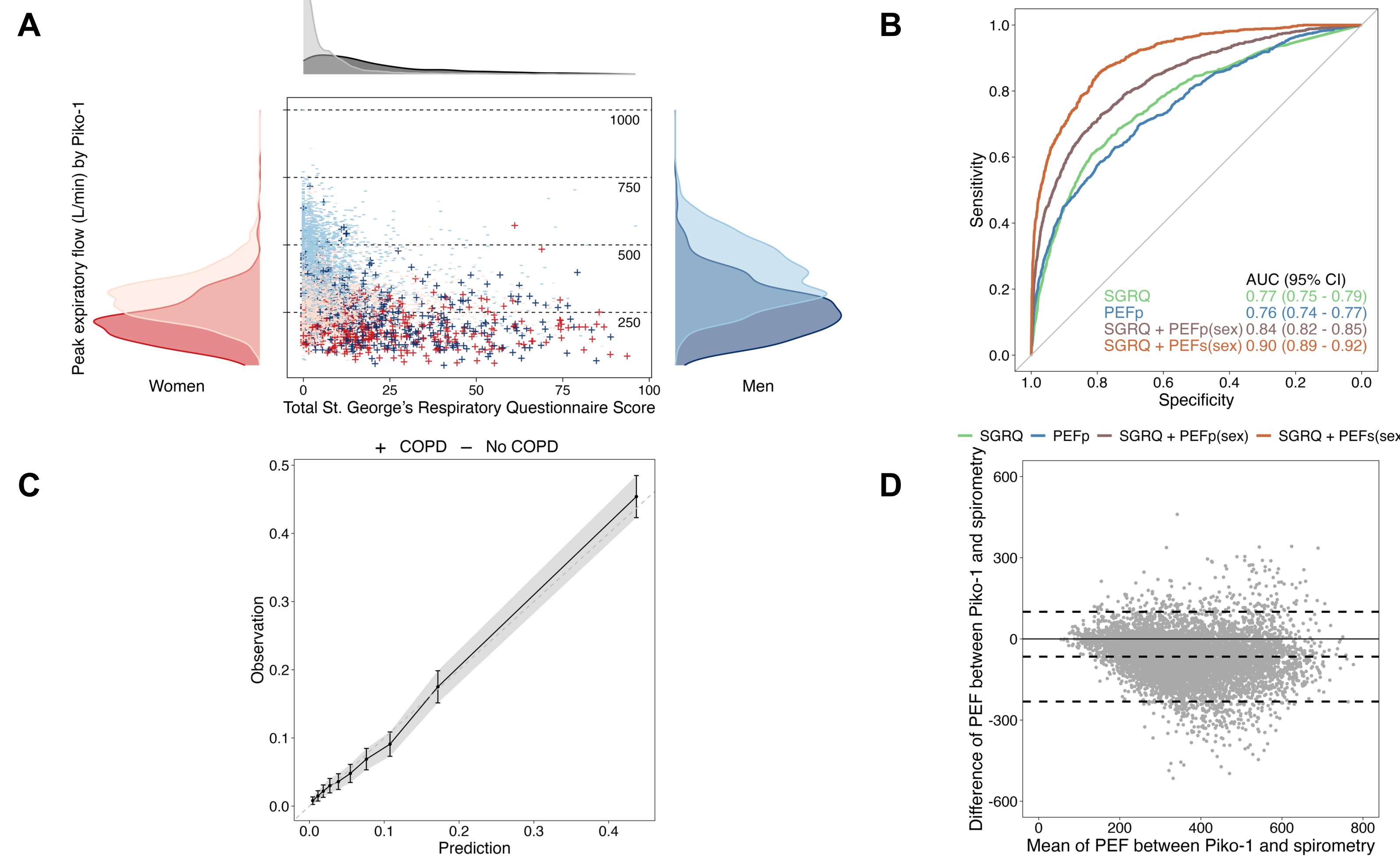
Figure 1. Association between age, sex, and COPD status on SGRQ score (A and B) and pre-bronchodilator PEF by peak flow meter (C and D), and association between SGRQ and lung function (E and F)



Results

Screening for COPD

- The AUC for SGRQ as a screening tool for COPD was 0.77 (95% CI 0.75-0.79) and best threshold was 10.75 points.
- When the SGRQ was combined with pre-bronchodilator PEF stratified by sex, AUC increased to 0.84 (95% CI 0.82-0.85).
- A screening test that combined a total SGRQ score ≥12 points and/or pre-bronchodilator PEF <400 L/min for men and <250 L/min for women yielded 91% sensitivity, 47% specificity and negative predictive value of 98% to identify spirometry-confirmed COPD.



Discussion

- SGRQ is a simple, inexpensive, and efficient screening tool for spirometry-confirmed COPD.
- Screening with SGRQ coupled with PEF may identify people at-risk for COPD in resource-limited settings where spirometry is limited.
- Further research is needed to test SGRQ implementation in other settings and if screening and earlier diagnosis is associated with better COPD outcomes.

Acknowledgements

We have no relevant disclosures to declare.

Funding: Supported by Medical Research Council (MR/P008984/1) and NIH Fogarty International Center (D43TW009340, D43TW011502).

References

- Chen S, Kuhn M, Prettner K, et al. The global economic burden of chronic obstructive pulmonary disease for 204 countries and territories in 2020-50: a health-augmented macroeconomic modelling study. *Lancet Glob Health*. 2023 Aug;11(8):e1183-e1193.
- Hurst JR, Buist AS, Gaga M, et al. Challenges in the Implementation of Chronic Obstructive Pulmonary Disease Guidelines in Low- and Middle-Income Countries: An Official American Thoracic Society Workshop Report. *Ann Am Thorac Soc*. 2021 Aug;18(8):1269-1277.
- Checkley W, Yang M, et al. **Population-based screening for chronic obstructive pulmonary disease using the St. George's Respiratory Questionnaire in resource-limited settings.** *Am J Respir Crit Care Med* 2025;211:779–788.
- Blanco I, Diego I, Bueno P, et al. Geographic distribution of COPD prevalence in the world displayed by Geographic Information System maps. *Eur Respir J*. 2019 Jul 18;54(1):1900610.